

30 Days SQL Micro course Answer PDF

Final Assignment:-

--Create Database

```
DROP DATABASE IF EXISTS Hospital;  
CREATE DATABASE Hospital;
```

--Create Tables

```
DROP TABLE IF EXISTS Hospital_Data;  
CREATE TABLE Hospital_Data(  
    Hospital_Name VARCHAR(100),  
    Location VARCHAR(80),  
    Department VARCHAR(100),  
    Doctors_Count INT,  
    Patients_COUNT INT,  
    Admission_Date DATE,  
    Discharge_Date DATE,  
    Medical_Expenses NUMERIC(10,2)  
);
```

```
SELECT*FROM Hospital_Data;]
```

-- Import Data into Hospital_Data Table

```
COPY Hospital_Data  
(Hospital_Name, Location, Department, Doctors_Count, Patients_Count, Admission_Date,  
Discharge_Date, Medical_Expenses)  
FROM 'C:/temp/Hospital_Data.csv'  
WITH (  
    FORMAT CSV,  
    HEADER TRUE,  
    DELIMITER ';' )
```

Questions:-

1. Total Number of Patients

- Write an SQL query to find the total number of patients across all hospitals.

Answer:- SELECT COUNT(*) AS Total_Number_of_patients
FROM Hospital_Data;

2. Average Number of Doctors per Hospital

- Retrieve the average count of doctors available in each hospital.

Answer:- SELECT Hospital_name,
AVG(Doctors_Count) AS average_doctors
FROM Hospital_data
GROUP BY Hospital_name;

3. Top 3 Departments with the Highest Number of Patients

- Find the top 3 hospital departments that have the highest number of patients.

Answer:- SELECT Department,
SUM(Patients_Count) AS total_patients
FROM hospital_data
GROUP BY Department
ORDER BY total_patients DESC
LIMIT 3;

4. Hospital with the Maximum Medical Expenses

- Identify the hospital that recorded the highest medical expenses.

Answer:- SELECT hospital_name,
MAX(medical_expenses) AS maximum_medical_expenses
FROM hospital_data
GROUP BY hospital_name
ORDER BY maximum_medical_expenses DESC
LIMIT 1;

5. Daily Average Medical Expenses

- Calculate the average medical expenses per day for each hospital.

Answer:- SELECT admission_date,
ROUND(AVG(medical_expenses), 2) AS daily_average_medical_expenses
FROM hospital_data
GROUP BY admission_date
ORDER BY admission_date;

6. Longest Hospital Stay

- Find the patient with the longest stay by calculating the difference between Discharge Date and Admission Date.

Answer:- SELECT hospital_name,
admission_date,
discharge_date,
(discharge_date - admission_date) AS hospital_stay_days
FROM hospital_data
ORDER BY hospital_stay_days DESC
LIMIT 1;

7. Total Patients Treated Per City

- Count the total number of patients treated in each city.

Answer:- SELECT location,
SUM(patients_count) AS total_patients
FROM hospital_data
GROUP BY location
ORDER BY total_patients DESC;

8. Average Length of Stay Per Department

- Calculate the average number of days patients spend in each department.

Answer:- SELECT department,
 ROUND(AVG(discharge_date - admission_date), 2) AS average_length_of_stay
FROM hospital_data
GROUP BY department
ORDER BY average_length_of_stay DESC;

9. Identify the Department with the Lowest Number of Patients

- Find the department with the least number of patients.

Answer:- SELECT department,
 SUM(patients_count) AS total_patients
FROM hospital_data
GROUP BY department
ORDER BY total_patients ASC
LIMIT 1;

10. Monthly Medical Expenses Report

- Group the data by month and calculate the total medical expenses for each month.

Answer:- SELECT TO_CHAR(admission_date, 'YYYY-MM') AS month,
 SUM(medical_expenses) AS total_medical_expenses
FROM hospital_data
GROUP BY TO_CHAR(admission_date, 'YYYY-MM')
ORDER BY month;